COMPUTER PROGRAMMING · PRACTICE

Sum and Difference

DESCRIPTION

A small display panel shows two integers, a on the left and b on the right. When the “update” button is pressed, both displays are rewritten at the same time:

- the left display shows the **sum** of the two old numbers, a + b;
- the right display shows how far apart the two old numbers were, that is, the **absolute difference** |a - b| (which is a - b if a ≥ b, and b - a otherwise, so it is never negative).

Both new values are calculated from the numbers that were on the display **before** the button was pressed. Given the two numbers, print what the display shows after one update.

REQUIRED FUNCTION

Your program must define the following function:

```
void update(int *a, int *b)
```

- `a` and `b` point to the two numbers on the display. The function replaces `*a` with the sum and `*b` with the absolute difference of the **original** values of `*a` and `*b`.
- It must read and change the numbers through `*a` and `*b`. It returns nothing.
- In `main`, read the two integers into two variables, call the function by passing their **addresses** with `&`, and then print the two variables.

Be careful with the order of your assignments: once `*a` has been overwritten, the old value of `*a` is gone.

INPUT FORMAT

The input contains one line with two integers a and b, separated by a single space.

$-10^6 \le a, b \le 10^6$. Either number may be negative or zero.

OUTPUT FORMAT

Print the two new numbers (the sum first, then the absolute difference) on one line, separated by a single space, followed by a newline.

SAMPLE

Sample 1 Input

```
7 3
```

Sample 1 Output

```
10 4
```

SOURCE

Adapted from HackerRank, “Pointers in C”.